

SE Lab 1 – Requirements Engineering

No 57 : Pet Adoption & Shelter Management System

Name : Harsha K

Section : 5K

SRN : PES1UG24CS666

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow shelter staff to log animal vaccination and medical intake records, updating the animal availability status to "Ready for Adoption".	High	Pass: A verified animal record contains the required vaccination/medical intake details and its status becomes "Ready for Adoption". Fail: An animal under medical quarantine is marked adopted.	Ensures medical records are maintained and only medically cleared animals become available for adoption.
FR-002	Functional	The system shall allow adoption applicants to search and filter available pets by species, age, breed, and whether the pet is good with kids.	High	Pass: Selecting any supported filter updates the result set to contain only matching available pets. Fail: Results contain an unavailable pet or a pet that violates a selected filter.	Helps applicants quickly identify pets that match their needs and household conditions.
FR-003	Functional	The system shall allow an adoption applicant to complete and submit a prospective-adopter questionnaire for a selected pet.	High	Pass: A completed questionnaire is successfully submitted and linked to the selected pet and applicant. Fail: Submission is accepted when any mandatory field is missing.	Provides the shelter with standardized information for evaluating prospective adopters.
FR-004	Functional	The system shall allow shelter staff to review submitted adoption applications and record an approve, reject, or request-more-information decision.	High	Pass: Staff can open a submitted application and save exactly one current decision with an optional note. Fail: An application cannot be saved without a decision.	Supports a traceable and consistent adoption screening process.
FR-005	Functional	The system shall allow shelter staff to create and update foster-home placement timelines for animals requiring foster care.	Medium	Pass: Staff can create a placement with start and expected end dates and later update its status. Fail: A placement with an end date earlier than its start date is accepted.	Supports temporary care planning and tracking before adoption or further shelter placement.
NFR-001	Nonfunctional	The pet search catalog must support multi-attribute filtering (Species, Age, Breed, Good with Kids) with a page response time of less than 200 ms.	High	Pass: Benchmarking tests under simulated peak load show the target response time of < 200 ms while applying the supported filters. Fail: The measured response time is 200 ms or greater.	Provides fast pet discovery and maintains responsiveness when many applicants use the catalog.
NFR-002	Nonfunctional	The system shall restrict access to medical intake records and adoption	High	Pass: An authorized staff user can access the protected records and an unauthorized	Protects sensitive animal medical and applicant information

		application information to authenticated users with the required role permissions.		user receives an access-denied response. Fail: An unauthorized user can view or modify protected records.	and prevents unauthorized access.
--	--	--	--	---	-----------------------------------

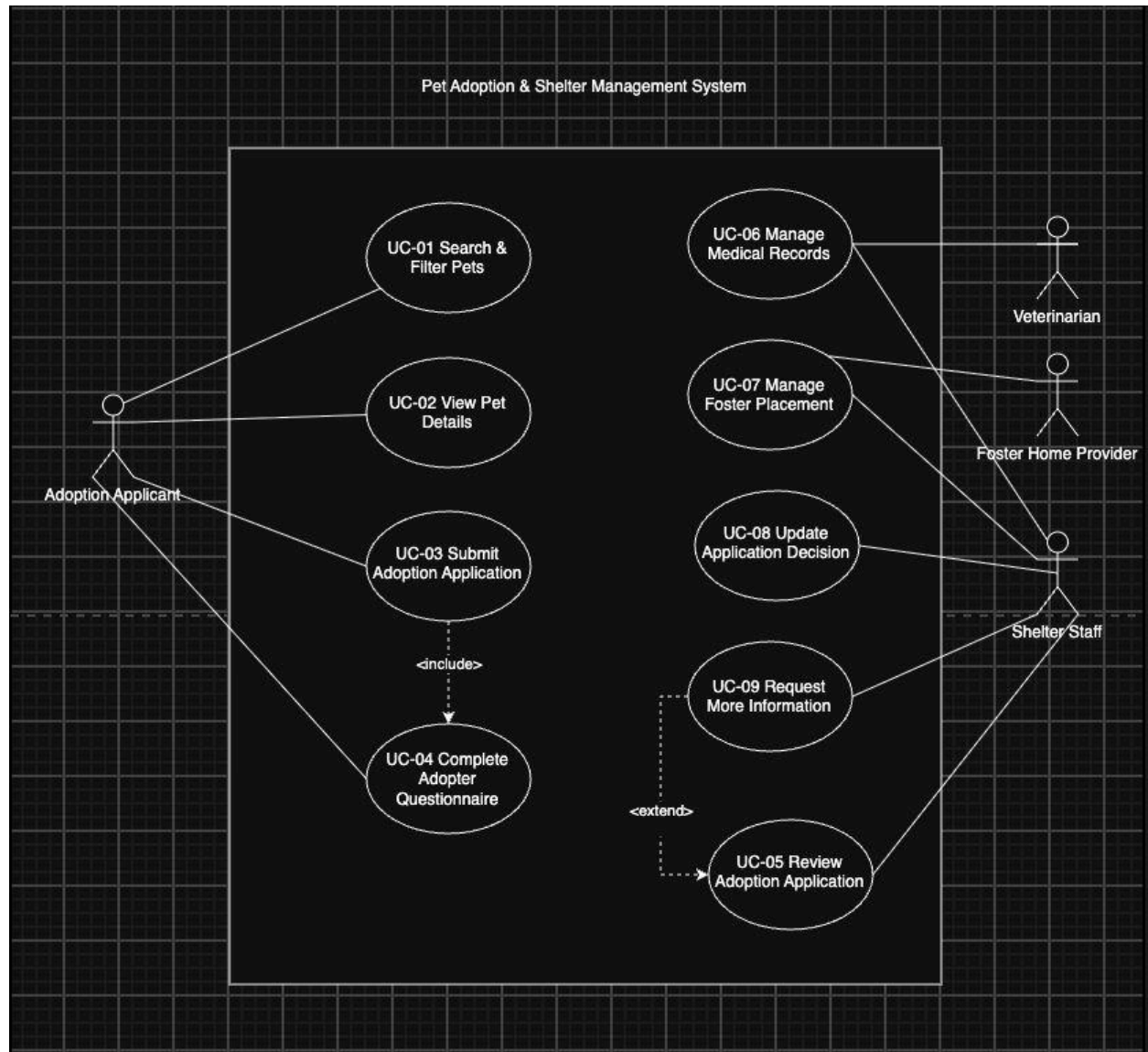
SE Lab 1 – Use-Case Diagram

No 57 : Pet Adoption & Shelter Management System

Name : Harsha K

SRN : PES1UG24CS666

Section : 5K



SE Lab 1 – Use-Case Flow Specification

No 57 : Pet Adoption & Shelter Management System

Name : Harsha K

SRN : PES1UG24CS666

Section : 5K

Use Case

UC-03 – Submit Adoption Application

Primary Actor

Adoption Applicant

Preconditions

- The adoption applicant is registered/logged into the system.
- The selected pet exists in the system and is currently available.
- The selected pet has the availability status “Ready for Adoption”.

Postconditions

Success: An adoption application is created, linked to the applicant and selected pet, the completed questionnaire is stored, and the application status is set to “Submitted”.

Failure: No incomplete or invalid application is submitted, and the applicant is informed about missing or invalid information.

Main Success Scenario

1. Adoption Applicant searches for an available pet.
2. The system displays matching available pets.
3. Applicant selects a pet.
4. The system displays the selected pet’s details.
5. Applicant selects “Apply for Adoption”.
6. The system displays the adopter questionnaire.
7. Applicants enter all required questionnaire information.
8. The system validates the questionnaire.
9. Applicant submits the adoption application.
10. The system creates the application and links it to the applicant and selected pet.
11. The system sets the application status to “Submitted”.
12. The system displays an application submission confirmation.
13. Use case ends successfully.

Alternate Flow – Missing Required Information

- A1.1 At Step 8, the system detects one or more missing mandatory fields.
- A1.2 System highlights the missing fields and displays a message requesting completion.
- A1.3 Applicant enters the missing information.
- A1.4 System validates the questionnaire again.
- A1.5 If the information is valid, the flow continues at Step 9 of the Main Success Scenario.